

“The changes in question **are different ... from any** that have been **before dealt with in chemistry.**”

Ernest Rutherford, 1902

Rutherford fired positively charged particles at a thin layer of atoms. Most of the particles went straight through the atoms, but some were deflected, and some bounced right back. He concluded that atoms consisted mainly of empty space with a tiny positively charged center, the nucleus, orbited by a cloud of electrons.

